

Special Function

Importance of the Gamma Function in AI & Data Science

- Generalizes the factorial: $\Gamma(n + 1) = n!$
- Appears in Gamma, Beta, Chi-square, t, and Dirichlet distributions
- Important in Bayesian machine learning
- Used in normalization of probability density functions
- Supports statistical inference, uncertainty modeling, and probabilistic AI

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Special functions are particular mathematical functions which have more or less established names and notations due to their importance in mathematical analysis, functional analysis, physics, or other applications.

Gamma function

Many important functions in applied sciences are defined via improper integrals. Maybe the most famous among them is the Gamma Function.

The gamma function is defined by

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt \quad (1)$$

Let $t = s^2$, then $dt = 2s ds$. So by substituting in eq. (1) we have

$$\Gamma(x) = 2 \int_0^{\infty} e^{-s^2} s^{2x-1} ds \quad (2)$$

EXAMPLE 1 ►

Prove that

$$(i) \quad \Gamma(1) = 1. \quad (ii) \quad \Gamma(2) = 1$$

SOLUTION

$$(i) \text{ Form eq. (1), } \Gamma(1) = \int_0^{\infty} e^{-t} dt = \left[-e^{-t} \right]_0^{\infty} = 1.$$

$$(ii) \quad \Gamma(2) = \int_0^{\infty} e^{-t} t dt \text{ (using integration by parts)}$$

$$u = t \quad , \quad dv = e^{-t} dt$$

$$du = dt \quad , \quad v = -e^{-t}$$

$$\Gamma(2) = \left[-te^{-t} \right]_0^{\infty} + \int_0^{\infty} e^{-t} dt = 0 + \left[-e^{-t} \right]_0^{\infty} = 1$$

EXAMPLE

Evaluate $\int_0^{\infty} e^{-x^2} x^{5.2} dx$, given that $(\Gamma(0.1) = 9.5135)$

SOLUTION

Comparing the integral with the integral (2)

$$5.2 = 2n - 1 \Rightarrow n = 3.1. \text{ So}$$

$$\begin{aligned} \int_0^{\infty} e^{-x^2} x^{5.2} dx &= \frac{1}{2} \Gamma(3.1) = \frac{1}{2} (2.1) \Gamma(2.1) \\ &= \frac{1}{2} (2.1)(1.1) \Gamma(1.1) \\ &= \frac{1}{2} (2.1)(1.1)(0.1) \Gamma(0.1) \\ &= \frac{1}{2} (2.1)(1.1)(0.1)(9.513) = 1.1 \end{aligned}$$

EXAMPLE

Find $\int_0^{\infty} (1 - 2x^2)^2 e^{-x} dx$

SOLUTION

$$\begin{aligned}
 I &= \int_0^{\infty} (1 - 2x^2)^2 e^{-x} dx \\
 &= \int_0^{\infty} (1 - 4x^2 + 4x^4) e^{-x} dx \\
 &= \int_0^{\infty} e^{-x} dx - 4 \int_0^{\infty} x^2 e^{-x} dx + 4 \int_0^{\infty} x^4 e^{-x} dx \\
 &= \Gamma(1) - 4\Gamma(3) + 4\Gamma(5) \\
 &= 1 - 4(2!) + 4(4!) = 89
 \end{aligned}$$

EXAMPLE

Calculate $\int_0^1 \frac{1}{\sqrt{-\ln x}} dx$

SOLUTION

Let $-\ln x = t \Rightarrow x = e^{-t} \Rightarrow dx = -e^{-t} dt$

Limits: as $x \rightarrow 0^+$, $t = -\ln x \rightarrow \infty$, $x = 1 \rightarrow t = 0$

Thus,

$$\int_0^1 \frac{1}{\sqrt{-\ln x}} dx = - \int_{\infty}^0 \frac{e^{-t}}{\sqrt{t}} dt = \int_0^{\infty} t^{-1/2} e^{-t} dt = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Solve

$$\int_0^3 \left(\ln\left(\frac{3}{x}\right)\right)^{1.5} dx$$

Let

$$t = \ln \frac{3}{x}$$

Now solve for x :

$$e^t = \frac{3}{x}$$

$$x = 3e^{-t}$$

Differentiate:

$$dx = -3e^{-t} dt$$

Change the limits

When $x = 3$, $t = \ln \frac{3}{3} = \ln 1 = 0$, When $x \rightarrow 0^+$, $\frac{3}{x} \rightarrow \infty \Rightarrow t = \ln \frac{3}{x} \rightarrow \infty$

So the limits become:

$$x: 0 \rightarrow 3 \Rightarrow t: \infty \rightarrow 0$$

$$dx = -3e^{-t} dt$$

then

$$I = \int_{\infty}^0 t^{3/2} (-3e^{-t}) dt$$

Reverse the limits:

$$I = 3 \int_0^{\infty} t^{3/2} e^{-t} dt$$

We use the standard formula

$$\Gamma(n) = \int_0^{\infty} t^{n-1} e^{-t} dt$$

Compare with

$$\int_0^{\infty} t^{3/2} e^{-t} dt$$

Here,

$$n - 1 = \frac{3}{2} \Rightarrow n = \frac{5}{2}$$

So

$$\int_0^{\infty} t^{3/2} e^{-t} dt = \Gamma\left(\frac{5}{2}\right)$$

Therefore,

$$I = 3\Gamma\left(\frac{5}{2}\right)$$

Use the Gamma-function property

$$\Gamma(n+1) = n\Gamma(n)$$

So,

$$\Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right)$$

and

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right)$$

Also,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Thus,

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}$$

and

$$\Gamma\left(\frac{5}{2}\right) = \frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi} = \frac{3}{4}\sqrt{\pi}$$

So,

$$I = 3 \cdot \frac{3}{4}\sqrt{\pi} = I = \frac{9}{4}\sqrt{\pi}$$

Evaluate:

$$\text{i) } \int_0^1 x^m (\ln x)^n dx, \quad m, n \text{ integers}$$

Solve by your hand very well like the following

$$\text{let } \ln x = -t \quad x = e^{-t}$$

$$dx = -e^{-t} dt$$

$$\text{as } x=0 \quad \ln 0 = -t \rightarrow -\infty = t$$

$$\text{so } t \rightarrow \infty$$

$$\text{as } x \rightarrow 1 \quad \ln 1 = -t \quad 0 = -t$$

$$t \rightarrow 0$$

$$I = \int_0^1 x^m (\ln x)^n dx = \int_0^1 (e^{-t})^m (-t)^n (-e^{-t} dt)$$

$$= (-1)^n \int_0^\infty e^{-tm} t^n e^{-t} dt = (-1)^n \int_0^\infty t^n e^{-(m+1)t} dt$$

$$(m+1)t = z \quad dt = \frac{dz}{m+1}$$

$$I = (-1)^n \int_0^\infty \left(\frac{z}{m+1}\right)^n e^{-z} \frac{dz}{m+1} = \frac{(-1)^n}{(m+1)^{n+1}} \int_0^\infty z^n e^{-z} dz$$

$$= \frac{(-1)^n}{(m+1)^{n+1}} \Gamma(n+1) = \frac{(-1)^n n!}{(m+1)^{n+1}}$$